Build an Obstacle-Avoiding Rover

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A Hands-On Workshop with Arduino, an Ultrasonic Sensor and a Servo

Participant Manual | Duration: about 3 hours | Level: Beginner

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1 Welcome and Goals

By the end of this workshop you will have built a small four-wheel (or two-wheel plus caster) rover that drives forward on its own, detects obstacles with an ultrasonic sensor, looks left and right with a servo, and chooses the clearer path.

What you will learn

- How a robot works as a loop of **sense** → **think** → **act**.
- How to drive DC motors with an **L298N** motor driver and PWM speed control.
- How to measure distance with an **HC-SR04** ultrasonic sensor.
- How to write small, testable pieces of code and combine them step by step.

Suggested schedule

Block	Activity	Time
1	Intro, how the rover thinks, parts check	20 min
2	Chassis assembly and motor mounting	30 min
3	Wiring (motors, driver, power)	30 min
4	Code Step 1: motor test	20 min
5	Wiring and Code Step 2: ultrasonic sensor	25 min
6	Mount servo, Code Step 3: full avoidance	35 min
7	Tuning, challenges, wrap-up	20 min

2 Parts List

Qty	Part	Notes
1	Arduino Uno (or compatible)	Nano also works; pin numbers are identical
1	L298N dual H-bridge motor driver	Red module with heatsink
1	2WD or 4WD robot chassis kit	Includes TT gear motors and wheels
1	HC-SR04 ultrasonic sensor	5 V version
1	SG90 micro servo	With a bracket or double-sided tape to hold the sensor
1	Battery holder, 2×18650 (7.4 V)	Or 6×AA (9 V)
1	Slide switch	To cut battery power
1	Jumper wires (M–M and M–F)	About 25
1	Small breadboard	Optional, helps share 5 V and GND
–	Screwdriver, tape, zip ties	For assembly

Tip

Use fully charged batteries. Weak batteries are the number one cause of "my rover resets randomly" problems.

3 How the Rover Thinks

The rover repeats the same loop many times per second:

Sense	→	**Think**	→	**Act**
Measure distance		Is something closer than 20 cm?		Drive, stop, turn

3.1 Ultrasonic distance explained

The HC-SR04 sends a short burst of 40 kHz sound from the `TRIG` pin trigger and listens for the echo. The `ECHO` pin stays HIGH for as long as the sound took to travel to the obstacle and back.

$$\text{distance (cm)} = \frac{\text{echo time } (\mu s)}{58}$$

The divisor 58 comes from the speed of sound (about 343 m/s), halved because the sound travels there and back.

3.2 Motor driver explained

The L298N is an H-bridge. For each motor it has two direction inputs and one enable input:

IN1	IN2	Motor A result
HIGH	LOW	Forward
LOW	HIGH	Backward
LOW	LOW	Stop (coast)
HIGH	HIGH	Brake

The enable pin (`ENA`) takes a PWM signal from `analogWrite()`, from 0 to 255, which sets the speed.

4 Assembly

1. Lay out the chassis plates and identify the top and bottom decks.
2. Attach the gear motors to the bottom plate with the supplied brackets and screws.
3. Solder or screw two wires to each motor terminal (if not pre-wired).
4. Press the wheels onto the motor shafts. For a 2WD kit, fit the caster wheel at the front or rear.
5. Mount the battery holder under or on top of the chassis.
6. Fix the L298N and the Arduino to the top deck using screws, standoffs or double-sided tape.
7. Mount the servo at the front edge of the top deck, and attach the HC-SR04 to the servo horn so it points forward when the servo is at 90°.

Tip

Before tightening anything, think about where the wires will run. Leave a little slack at the servo so the sensor can swing left and right freely.

5 Wiring

Warning

Always wire with the battery **disconnected**. Double-check each connection before switching on. A swapped power wire can destroy the Arduino.

5.1 Motors and driver

L298N pin	Connects to	Purpose
OUT1, OUT2	Left motor(s)	Left side
OUT3, OUT4	Right motor(s)	Right side
ENA	Arduino D5	Left speed (PWM)
IN1	Arduino D7	Left direction
IN2	Arduino D8	Left direction
IN3	Arduino D9	Right direction
IN4	Arduino D10	Right direction
ENB	Arduino D6	Right speed (PWM)

Remove the jumpers on `ENA` and `ENB` so that the Arduino can control them. On a 4WD kit, wire the two left motors in parallel to OUT1/OUT2 and the two right motors in parallel to OUT3/OUT4.

5.2 Sensor and servo

Part pin	Connects to	Note
HC-SR04 VCC	Arduino 5 V	
HC-SR04 GND	Arduino GND	
HC-SR04 TRIG	Arduino D12	Output from Arduino
HC-SR04 ECHO	Arduino D11	Input to Arduino
Servo red	Arduino 5 V	
Servo brown/black	Arduino GND	
Servo orange/yellow	Arduino D3	Signal

5.3 Power

From	To	Note
Battery + (via switch)	L298N 12 V terminal	Motor power
Battery −	L298N GND terminal	
L298N GND	Arduino GND	**Common ground is essential**
L298N 12 V terminal	Arduino VIN	7.4 V is fine for VIN

Warning

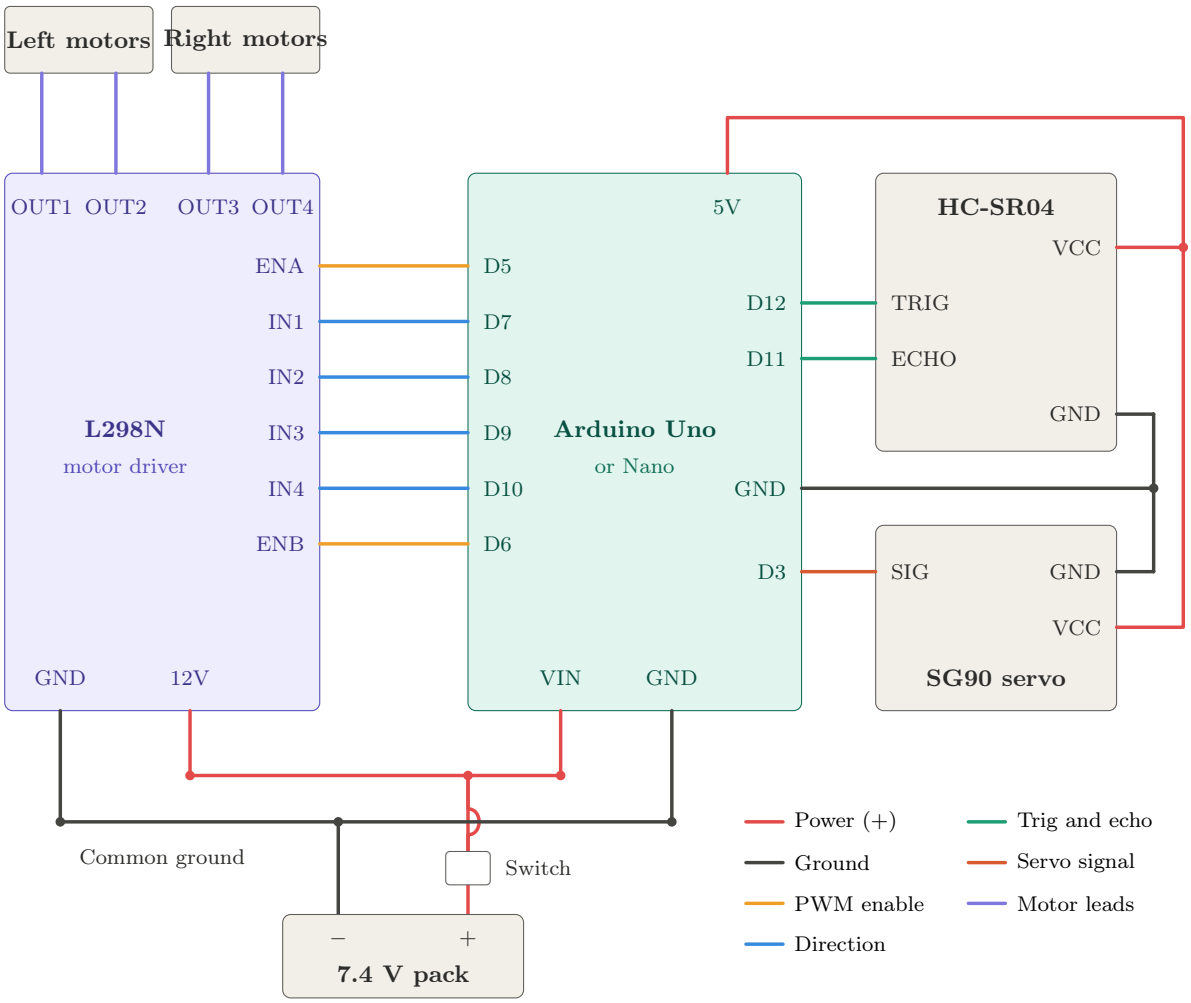
Do not connect a USB cable and the battery to the Arduino at the same time while the motors are running. Use USB only for uploading code, with the battery switch off.

Checkpoint

Before continuing: every GND is joined together, nothing is connected to the wrong voltage, and the battery switch is off.

5.4 Complete circuit diagram

The diagram below combines every connection from the tables above. Use it to double-check your wiring before switching on.



Complete wiring diagram. The red wire hops over the ground line without connecting to it; dots mark real junctions.

6 Code

Install the Arduino IDE from `https://www.arduino.cc/en/software`. The `Servo` library is already included. Select **Tools** → **Board** → **Arduino Uno** and the correct port before uploading.

Build in three small steps, so that if something breaks you know exactly where.

6.1 Step 1: Motor test

Lift the rover so the wheels spin in the air. Upload and watch.

```
// STEP 1: Motor test - each motion runs for 2 seconds
const uint8_t ENA = 5, IN1 = 7, IN2 = 8;    // left motor
const uint8_t ENB = 6, IN3 = 9, IN4 = 10;   // right motor
const uint8_t SPEED = 150;                  // 0 to 255

void setup() {
  pinMode(ENA, OUTPUT); pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
  pinMode(ENB, OUTPUT); pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
}

void drive(bool lFwd, bool rFwd) {
  digitalWrite(IN1, lFwd);  digitalWrite(IN2, !lFwd);
  digitalWrite(IN3, rFwd);  digitalWrite(IN4, !rFwd);
```

```
  analogWrite(ENA, SPEED);  analogWrite(ENB, SPEED);
}

void stopMotors() {
  analogWrite(ENA, 0);  analogWrite(ENB, 0);
}

void loop() {
  drive(true, true);   delay(2000);   // forward
  stopMotors();        delay(1000);
  drive(false, false); delay(2000);   // backward
  stopMotors();        delay(1000);
  drive(false, true);  delay(2000);   // spin left
  stopMotors();        delay(1000);
  drive(true, false);  delay(2000);   // spin right
  stopMotors();        delay(1000);
}
```

Checkpoint

Forward should make *both* sides turn forward. If one side runs backward, swap that motor's two wires on the L298N output. If "spin left" actually spins right, swap the left and right motor pairs.

6.2 Step 2: Ultrasonic sensor test

Open the Serial Monitor at 9600 baud and wave your hand in front of the sensor.

```
// STEP 2: Distance sensor test
const uint8_t TRIG = 12, ECHO = 11;

void setup() {
  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);
  Serial.begin(9600);
}

long readCm() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  long us = pulseIn(ECHO, HIGH, 30000);   // timeout 30 ms (about 5 m)
  if (us == 0) return 400;                // nothing detected
  return us / 58;
}

void loop() {
  Serial.print("Distance: ");
  Serial.print(readCm());
  Serial.println(" cm");
  delay(200);
}
```

Checkpoint

The reading should drop as your hand approaches and stay stable at a fixed distance. Check it against a ruler at 10 cm and 30 cm.

6.3 Step 3: Full obstacle-avoiding rover

This combines the motors, sensor and servo. The rover drives forward until an obstacle is closer than `STOP_CM`, then it stops, backs up briefly, looks left and right, and turns toward the side with more free space.

```
// STEP 3: Obstacle-avoiding rover
#include <Servo.h>

// ---- Pins ----
const uint8_t ENA = 5, IN1 = 7, IN2 = 8;      // left motor
const uint8_t ENB = 6, IN3 = 9, IN4 = 10;     // right motor
const uint8_t TRIG = 12, ECHO = 11;           // ultrasonic
const uint8_t SERVO_PIN = 3;

// ---- Tunable settings ----
const int  CRUISE_SPEED = 150;   // forward speed (0-255)
const int  TURN_SPEED   = 170;   // spinning speed
const int  STOP_CM      = 20;    // react when closer than this
const int  TURN_MS_90   = 450;   // time to turn about 90 degrees (calibrate!)

const int  LOOK_LEFT = 160, LOOK_CENTER = 90, LOOK_RIGHT = 20;

Servo head;

// ---------- Sensing ----------
long readCm() {
  digitalWrite(TRIG, LOW);  delayMicroseconds(2);
  digitalWrite(TRIG, HIGH); delayMicroseconds(10);
  digitalWrite(TRIG, LOW);
  long us = pulseIn(ECHO, HIGH, 30000);
  if (us == 0) return 400;
  return us / 58;
}

// Average of 3 readings to reduce noise
long readCmSmooth() {
  long sum = 0;
  for (int i = 0; i < 3; i++) { sum += readCm(); delay(10); }
  return sum / 3;
}

long lookAt(int angle) {
  head.write(angle);
  delay(450);                    // give the servo time to arrive
  return readCmSmooth();
}

// ---------- Motion ----------
// speed: -255 (full reverse) to +255 (full forward)
void setMotor(uint8_t en, uint8_t inA, uint8_t inB, int speed) {
  digitalWrite(inA, speed > 0);
  digitalWrite(inB, speed < 0);
  analogWrite(en, abs(speed));
}

void drive(int left, int right) {
  setMotor(ENA, IN1, IN2, left);
  setMotor(ENB, IN3, IN4, right);
}

void stopMotors()    { drive(0, 0); }
void forward()       { drive(CRUISE_SPEED, CRUISE_SPEED); }
void backward()      { drive(-CRUISE_SPEED, -CRUISE_SPEED); }
void spinLeft()      { drive(-TURN_SPEED, TURN_SPEED); }
```

```
void spinRight()     { drive(TURN_SPEED, -TURN_SPEED); }

// ---------- Setup ----------
void setup() {
  pinMode(ENA, OUTPUT); pinMode(IN1, OUTPUT); pinMode(IN2, OUTPUT);
  pinMode(ENB, OUTPUT); pinMode(IN3, OUTPUT); pinMode(IN4, OUTPUT);
  pinMode(TRIG, OUTPUT); pinMode(ECHO, INPUT);

  head.attach(SERVO_PIN);
  head.write(LOOK_CENTER);
  Serial.begin(9600);
  delay(2000);                    // 2 s to put the rover on the floor
}

// ---------- Main loop ----------
void loop() {
  long ahead = readCmSmooth();
  Serial.print("Ahead: "); Serial.println(ahead);

  if (ahead > STOP_CM) {
    forward();
    delay(50);
    return;
  }

  // Obstacle detected
  stopMotors();
  delay(200);
  backward();  delay(300);
  stopMotors();

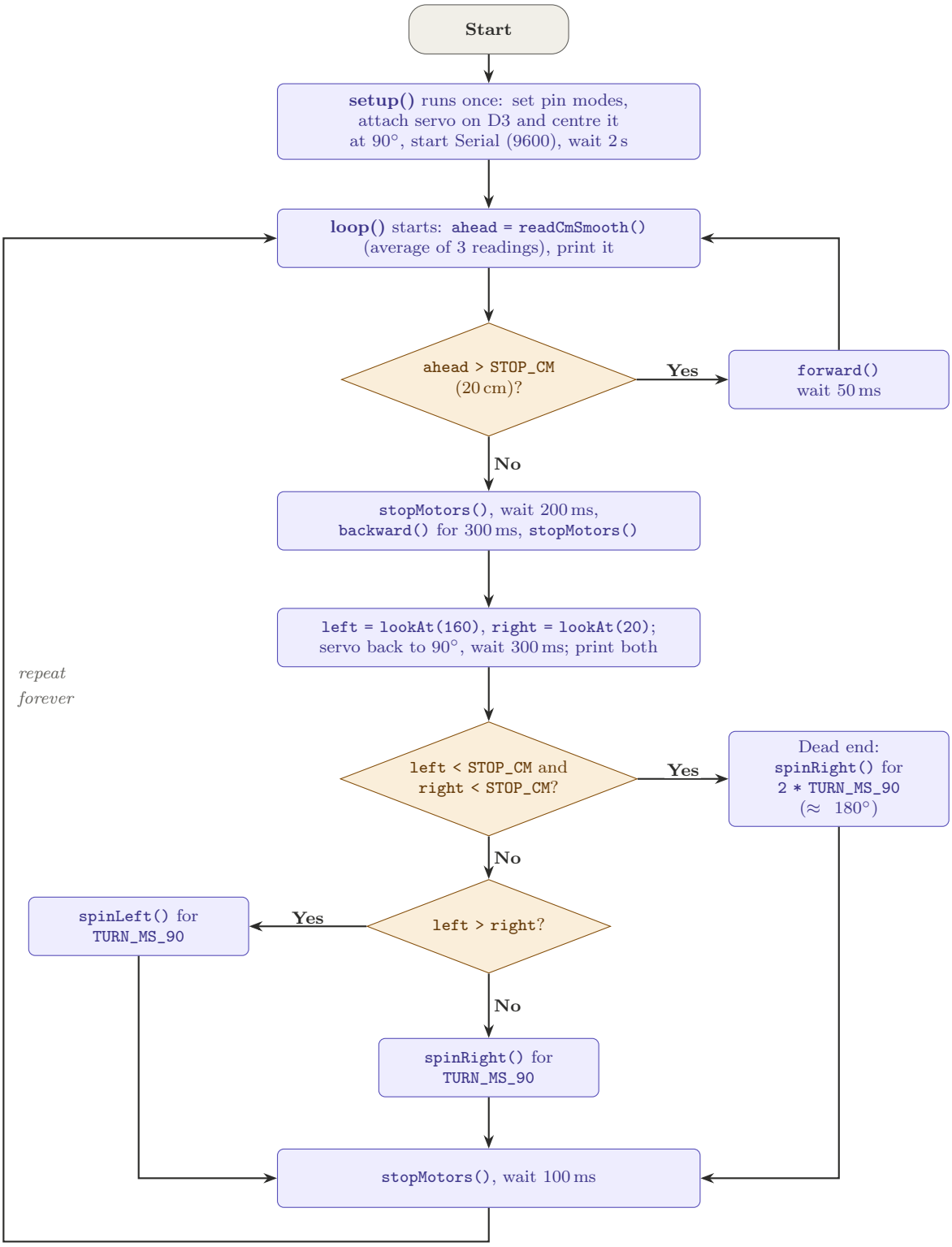
  long left  = lookAt(LOOK_LEFT);
  long right = lookAt(LOOK_RIGHT);
  head.write(LOOK_CENTER);
  delay(300);

  Serial.print("Left: ");  Serial.print(left);
  Serial.print("  Right: "); Serial.println(right);

  if (left < STOP_CM && right < STOP_CM) {
    spinRight();                  // dead end: turn around (about 180 degrees)
    delay(TURN_MS_90 * 2);
  } else if (left > right) {
    spinLeft();
    delay(TURN_MS_90);
  } else {
    spinRight();
    delay(TURN_MS_90);
  }
  stopMotors();
  delay(100);
}
```

6.4 Program flowchart

The flowchart shows the logic of the Step 3 program: `setup()` runs once, then `loop()` repeats forever.



Flowchart of the Step 3 program. `STOP_CM` = 20 cm and `TURN_MS_90` = 450 ms by default.

6.5 How the code works

`readCm()`
: Fires one ultrasonic pulse and converts the echo time to centimetres.

`readCmSmooth()`
: Averages three readings so a single bad echo does not trigger a false stop.

`setMotor()`
: Takes a signed speed; the sign picks the direction pins and the magnitude drives PWM.

`loop()`
: Drives forward while the path is clear; otherwise stop, back up, scan, and turn toward the larger gap.

7 Tuning and Calibration

Setting	What to adjust
`STOP_CM`	Raise to 30 for a faster rover that needs more braking room. Lower to 15 for tight spaces.
`CRUISE_SPEED`	Lower (about 120) for more accurate sensing; higher for speed. Below about 90, most TT motors stall.
`TURN_MS_90`	Place the rover on the floor, trigger a turn and measure the angle. Increase or decrease until it turns about 90°. This depends on battery level and floor surface.
Servo angles	If the sensor does not point straight ahead at 90, loosen the horn and re-attach it, or add an offset in code.

8 Troubleshooting

Problem	Likely fix
Nothing happens at all	Check the battery switch and charge level. Confirm the L298N power LED is on.
Arduino resets when motors start	Weak batteries or a loose power wire. Use fresh cells. Add a 470 μF capacitor across the battery terminals.
Motors hum but do not turn	Speed too low. Raise `CRUISE_SPEED` to 150 or more.
One side never moves	Check that the `ENA`/`ENB` jumper is removed and the enable wire reaches D5/D6. Check the motor wires.
Rover drives backward in forward mode	Swap the motor wires on both output pairs, or swap `IN1`/`IN2` and `IN3`/`IN4` in code.
Distance always reads 400	TRIG and ECHO swapped, or 5 V/GND loose on the sensor.
Distance jumps randomly	Soft surfaces (cloth) and angled walls reflect sound poorly. Test against a flat hard surface.
Servo jitters or twitches	Servo is browning out. Power it from a separate 5 V supply with a common ground.
Rover hits low obstacles	The sensor beam is cone shaped and sits high. Tilt it slightly down or add a bumper.

9 Challenges and Extensions

1. **Speed up.** Make the rover slow down smoothly as it approaches an obstacle instead of stopping suddenly.
2. **Wall follower.** Use the servo at 20° to hold a fixed distance to a wall.
3. **Status LEDs and buzzer.** Beep when backing up and light an LED when turning.
4. **Bluetooth override.** Add an HC-05 module and drive manually from your phone.
5. **Maze run.** Build a cardboard maze and time how fast your rover escapes.
6. **Mapping.** Print the scan angles and distances over Serial and plot them to see the room outline.

10 Facilitator Notes

- Pre-check every kit: motors spin, servo moves, sensor reads, batteries charged.
- Pre-solder motor leads to save about 15 minutes.
- Work in pairs: one person wires, the other checks against the tables, then swap.
- Mark a start line and a test arena of about $2\,\text{m} \times 2\,\text{m}$ with cardboard walls.
- Keep spare L298N modules and batteries; they are the most common failures.
- Encourage participants to change one setting at a time and observe.

Happy building! Take pictures, share your results, and keep improving your rover.